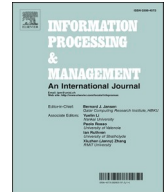




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SocLitGen: An LLM-assisted framework for automated literature review in social sciences

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ABSTRACT

Existing automated literature review technologies primarily target computer science contexts and face significant adaptability challenges in social science disciplines. Addressing this gap, this study proposes SocLitGen, a seven-module framework specifically designed for social science literature review generation. SocLitGen was evaluated using review tasks from management and economics topics on a corpus of 1,559,141 papers, comparing against Naive RAG, AutoSurvey, and graduate student writing. Baseline comparisons show that in citation quality (recall and precision), SocLitGen outperforms Naive RAG by at least 39.9%; in content quality (academic professionalism, structural rationality, and language fluency), SocLitGen outperforms AutoSurvey by at least 13.1% and graduate student writing by at least 9.5%. Cross-context evaluations reveal differentiated performance patterns: cross-disciplinary analysis confirms consistent performance across management and economics; cross-model experiments show Qwen2.5 performs optimally in academic and structural dimensions while GLM-4 excels in language fluency; cross-literature scale analysis demonstrates that large-scale sets enhance professionalism and structure but reduce fluency.

1. Introduction

Social science researchers face unprecedented challenges in conducting literature reviews due to the exponential growth of scientific publications. A literature review—a scholarly process of systematically identifying, evaluating, and synthesizing existing research (Ermel et al., 2021)—plays a vital role in bridging prior knowledge with new inquiries by revealing knowledge gaps, integrating relevant findings, and offering a foundation for developing research questions and hypotheses (Snyder, 2019). However, this fundamental scholarly practice has become increasingly burdensome as scientific literature has grown at an annual rate of 5.08 % from 1952 to 2020, with the volume approximately doubling every 14 years (Bornmann et al., 2021). This exponential knowledge expansion places enormous pressure on researchers who must navigate through an ever-expanding ocean of publications to identify pertinent studies, while simultaneously ensuring accurate comprehension and effective synthesis of complex information (Lok, 2010). The burden becomes particularly pronounced for social science researchers, as publication volumes in social science disciplines consistently surpass those of other fields by at least 13.3 % (Abt & Garfield, 2002). Given these mounting pressures, exploring technologies to alleviate the burden of social science literature reviews has become critically important for advancing scholarly productivity and research quality.

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Recent advances in Large Language Models (LLMs) offer new opportunities to alleviate these burdens and transform how researchers conduct literature reviews. LLMs are typically built upon Transformer architectures and trained on massive textual corpora (Vaswani et al., 2017). Through this training process, LLMs have attained human-level—or in some cases, superhuman—language comprehension and generation capabilities (Aher et al., 2023; Gilardi et al., 2023; Grossmann et al., 2023; Kosinski, 2023; Salvi et al., 2025). An increasing number of researchers are incorporating LLMs into their scientific writing workflows (Huang et al., 2025a; Prillaman, 2024). While generic LLMs alone remain insufficient for end-to-end literature review generation (Zimmermann et al., 2024), many researchers have developed systematic frameworks based on LLMs to facilitate literature screening (Joos et al., 2025; Long et al., 2025) understanding (Li & Xu, 2023; Li et al., 2023), evaluation (Huang et al., 2025b), and synthesis (Chung et al., 2025; Wang et al., 2023), demonstrating LLMs' remarkable potential in this domain. These successful applications across various literature review stages suggest that LLMs could serve as powerful tools for comprehensive literature review automation, offering unprecedented opportunities to alleviate the burden of literature synthesis while maintaining the rigor and depth required for scholarly work (Xu et al., 2024).

Building upon these promising developments in specialized applications, researchers have further advanced to propose comprehensive automated literature review frameworks that enable LLMs to undertake complete literature review tasks through systematic pipeline designs, with most efforts concentrated on computer science literature reviews. Notable examples include TAG (Chen et al., 2022), UR3WG (Shi et al., 2023), SLRs (Sami et al., 2024), LitLLM (Agarwal et al., 2024), RefAI (Li et al., 2024), ScholaCite (Martin-Boyle et al., 2024), AutoSurvey (Wang et al., 2024), Citegeist (Beger & Henneking, 2025), SurveyX (Liang et al., 2025), SurveyGO (Wang et al., 2025), and AutomaticReviewGeneration (Wu et al., 2025). These comprehensive frameworks have demonstrated considerable success, establishing the technical feasibility of end-to-end LLM-based literature review automation.

However, there is a notable absence of specialized frameworks designed specifically for social science literature review generation, and existing automated frameworks face significant adaptability challenges when applied to social science contexts. Specifically, current frameworks were developed primarily for computer science literature reviews (Agarwal et al., 2024; Beger & Henneking, 2025; Chen et al., 2022; Hoang & Kan, 2010; Hu & Wan, 2014; Liang et al., 2025; Martin-Boyle et al., 2024; Sami et al., 2024; Shi et al., 2023; Wang et al., 2025; Wang et al., 2020; Wang et al., 2024). When transferring these frameworks to social science contexts, they encounter significant adaptability challenges due to fundamental disciplinary differences across two critical dimensions. First, in terms of literature comprehension, social science research focuses on examining relationships between variables (Hofman et al., 2017), requiring scholars to extract theories, methodological approaches, samples and data, and empirical findings used to test these variable relationships from each paper. In contrast, computer science research concentrates on technical approaches for accomplishing computational tasks (Hofman et al., 2021), leading scholars to primarily extract task definitions and technical solutions. Second, in terms of review organization, social science scholars organize literature around competing theoretical perspectives to map the conceptual landscape of different explanations for phenomena, while computer science scholars organize around technological evolution to trace the developmental trajectory of technical approaches over time (e.g., LeCun et al., 2015). These fundamental differences suggest that frameworks designed for computer science may inadequately serve the distinct needs of social science literature synthesis.

Given these mounting pressures and the limitations of existing automated frameworks, this study aims to design an effective automated literature review solution specifically tailored for social sciences. The subsequent sections of this paper are organized as follows: Section 2 presents the research questions and objectives guiding this investigation. Section 3 provides a comprehensive review of existing automated literature review frameworks, systematically examining the developmental trajectory and core methodologies of current technical approaches. Section 4 presents a detailed exposition of SocLitGen, the automated framework specifically designed for social science literature reviews proposed in this study, including framework architecture design, core module functionality, and technical implementation pathways. Section 5 describes the experimental design covering dataset construction, evaluation metrics, and baseline selection, and reports results across baseline comparisons and cross-context evaluations. Section 6 discusses the findings in relation to the two research questions. Section 7 elaborates on the theoretical contributions and practical implications of this research. Section 8 concludes the paper by summarizing the main findings, candidly discussing the limitations of the current framework, and outlining promising directions for future research.

2. Research objectives

The general objective of this study is to propose an automated literature review framework tailored for social sciences. This framework specifically adapts literature comprehension and review organization logic to meet the requirements of social science scholarship. For such a framework to be practically viable, two dimensions must be addressed: first, the framework must demonstrate superior performance in literature review generation; second, the performance boundaries of the framework must be clearly established. Accordingly, this study formulates two research questions to guide the subsequent investigation.

- **RQ1:** How does the proposed framework perform in social science literature review generation compared to existing approaches?
To address RQ1, this study sets its first research objective as constructing a large-scale social science literature database, developing a comprehensive evaluation framework for assessing literature review quality, and applying the proposed framework to real-world social science literature review tasks. The framework's performance is then benchmarked against existing approaches, including human writing baselines and representative automated frameworks from other domains.
- **RQ2:** How does the performance of the proposed framework vary across different task contexts?

To address RQ2, this study sets its second research objective as systematically evaluating the framework's performance boundaries across three critical contexts: cross-disciplinary evaluation tests performance consistency across different social science

fields; cross-model evaluation examines how different underlying LLMs affect output quality and establishes optimal configurations; and cross-literature scale evaluation tests performance variation across different input sizes to establish the boundaries of effective automated synthesis.

The two RQs directly inform the study's methodological approach and experimental design. RQ1 structures the experimental evaluation framework, including baseline comparisons with human writing and existing automated approaches, while RQ2 guides the cross-context evaluation design, encompassing cross-disciplinary validation, cross-model configuration testing, and cross-literature scale analysis.

3. Related work

This section systematically examines existing automated literature review generation frameworks from a developmental perspective to understand their technical evolution and optimization trajectories. Current research in automated literature review generation has undergone three distinct developmental stages (see Fig. 1), each characterized by specific technological capabilities and limitations across three core technical dimensions: *literature acquisition*, *sub-topic planning*, and *review generation*. The following analysis traces this evolutionary path to illuminate the progressive optimization of automated literature review technologies.

The initial stage of automated literature review generation research, spanning before 2022, includes representative frameworks such as ReWoS, ARWG, RWS-Cit, and ToC-RWG (Chen & Zhuge, 2019; Hoang & Kan, 2010; Hu & Wan, 2014; Wang et al., 2020). These frameworks demonstrate three key characteristics: First, regarding *literature acquisition*, early frameworks exhibited weak literature retrieval capabilities by employing simple keyword-based text retrieval or requiring manual document provision. For example, the ReWoS framework proposed by Hoang and Kan (2010) for computer science literature review generation adopted basic keyword-based retrieval methods without advanced ranking algorithms. Moving toward a more simplified literature acquisition strategy, the ToC-RWG framework developed by Wang et al. (2020) for computer science literature review generation demonstrated this limitation by requiring users to manually provide target papers and reference literature rather than employing intelligent search mechanisms. Second, in terms of *sub-topic planning*, this stage was characterized by weak sub-topic planning capabilities through manual topic specification, basic unsupervised clustering methods, or complete avoidance of sub-topic planning. The ReWoS framework exemplified the first approach by requiring manual specification of review topics and sub-topic lists for structural planning without intelligent automation (Hoang & Kan, 2010). Advancing toward computational automation to reduce manual intervention, the ARWG framework introduced by Hu and Wan (2014) for computer science literature review generation relied on basic unsupervised methods for sub-topic construction based on input papers and reference literature, employing simple clustering algorithms for topic discovery. Moving toward an even more simplified approach that bypassed planning complexity entirely, the RWS-Cit framework developed by Chen and Zhuge (2019) for computational linguistics literature review generation demonstrated another strategy by proceeding without explicit sub-topic planning, processing literature directly without thematic organization. Third, regarding *review generation*, these frameworks employed text extraction and concatenation methods rather than sophisticated content synthesis. Indeed, the four frameworks—ReWoS, ARWG, RWS-Cit, and ToC-RWG—all employed the same crude approach of extracting and filtering sentences according to topical or sub-topic relevance and concatenating them to form reviews, highlighting the universal reliance on basic text assembly methods without advanced synthesis capabilities during this foundational period (Chen & Zhuge, 2019; Hoang & Kan, 2010; Hu & Wan, 2014; Wang et al., 2020).

The second developmental stage, spanning from 2022 to 2024, includes representative frameworks such as TAG, UR3WG, SLRs, LitLLM, and RefAI (Agarwal et al., 2024; Chen et al., 2022; Li et al., 2024; Sami et al., 2024; Shi et al., 2023). Compared to the first stage, these frameworks demonstrate significant technological improvements in two dimensions: First, regarding *literature acquisition*. Consider, for instance, the UR3WG framework presented by Shi et al. (2023) for computer science literature review generation demonstrated hybrid retrieval strategies that combined textual and semantic matching for sophisticated literature ranking. Advancing with AI-assisted refinement, the SLRs framework by Sami et al. (2024) for computer science literature review generation illustrated the integration of LLM-assisted literature judgment and screening processes for improved quality control. Expanding to multi-input strategies, the LitLLM framework proposed by Agarwal et al. (2024) for computer science literature review generation showcased comprehensive search approaches by extracting terms from keywords, abstracts, and related papers simultaneously. Moving toward metadata-based systems, the RefAI framework by Li et al. (2024) for biomedical literature review generation additionally implemented post-retrieval similar article recommendations based on literature metadata ranking algorithms. Second, in terms of *sub-topic planning*, this stage showed no significant advancement, with frameworks continuing to exhibit weak sub-topic planning capabilities. For example, the TAG, UR3WG, SLRs, LitLLM, and RefAI frameworks all bypassed explicit sub-topic planning, proceeding directly from literature acquisition to review generation without intelligent structural organization (Agarwal et al., 2024; Chen et al., 2022; Li et al., 2024; Sami et al., 2024; Shi et al., 2023). Third, regarding *review generation*, frameworks achieved enhanced generative summarization capabilities through the adoption of LLMs. Most notably, the TAG framework introduced by Chen et al. (2022) for computer science literature review generation marked this revolutionary transition by pioneering the use of LLM-based generative models for intelligent content synthesis rather than simple text assembly, establishing generative summarization as the new paradigm for automated review creation. The LitLLM, RefAI, and other frameworks of this period similarly adopted LLM-based approaches for review generation,

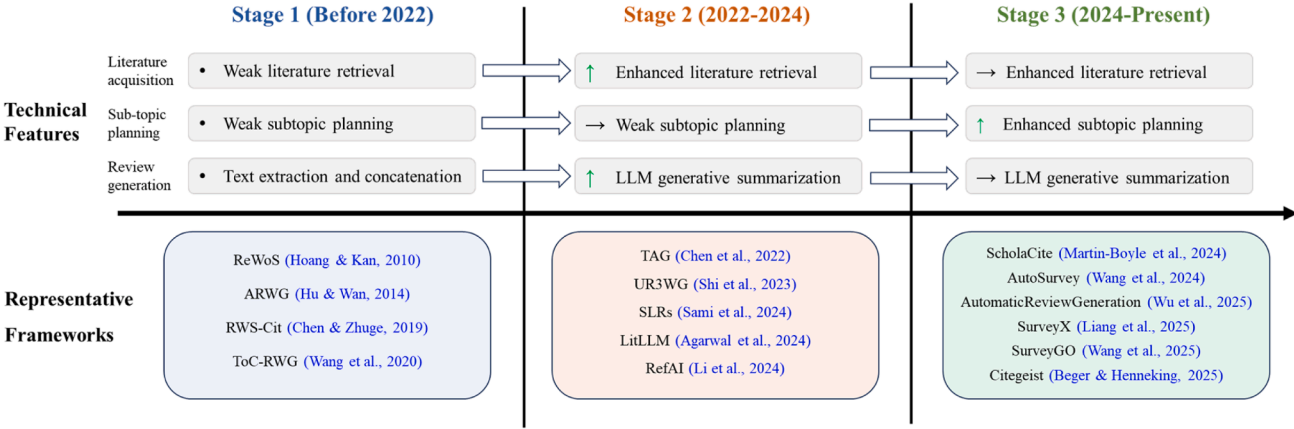


Fig. 1. Literature review automation technical evolution timeline.

collectively demonstrating the technological leap from text extraction to intelligent content synthesis (Agarwal et al., 2024; Li et al., 2024; Sami et al., 2024; Shi et al., 2023).

The third and current stage, beginning in 2024, includes representative frameworks such as ScholaCite, AutomaticReviewGeneration, AutoSurvey, Citegeist, SurveyX, and SurveyGO (Beger & Henneking, 2025; Liang et al., 2025; Martin-Boyle et al., 2024; Wang et al., 2025; Wang et al., 2024; Wu et al., 2025). These frameworks demonstrate three key characteristics based on technical feature evolution: First, regarding *literature acquisition*, frameworks maintained enhanced literature retrieval capabilities by continuing the advances from stage two. For example, the AutomaticReviewGeneration framework proposed by Wu et al. (2025) for chemistry literature review generation and the SurveyX framework by Liang et al. (2025) for computer science literature review generation adopted LLM-assisted literature judgment and screening processes, while the AutoSurvey framework developed by Wang et al. (2024) for computer science literature review generation and the Citegeist framework introduced by Beger and Henneking (2025) for computer science literature review generation implemented hybrid retrieval strategies combining textual and vector-based approaches. Second, in terms of *sub-topic planning*, this stage achieved enhanced sub-topic planning capabilities, representing a revolutionary breakthrough from the weak planning methods of previous stages. For example, the AutomaticReviewGeneration framework demonstrated LLM-based sub-topic generation by mimicking existing literature reviews to create structured sub-topic lists (Wu et al., 2025). The ScholaCite, AutoSurvey, SurveyX, and SurveyGO frameworks adopted reference-based planning approaches, using LLMs to generate structured outlines and sub-topic lists based on reference literature (Liang et al., 2025; Martin-Boyle et al., 2024; Wang et al., 2025; Wang et al., 2024). Third, regarding *review generation*, all frameworks in this stage adopted LLM-based generative summarization to organize and generate reviews, building upon the breakthrough established in the second stage to achieve sophisticated content synthesis capabilities across all systems (Beger & Henneking, 2025; Liang et al., 2025; Martin-Boyle et al., 2024; Wang et al., 2025; Wang et al., 2024; Wu et al., 2025).

This developmental trajectory reveals a clear optimization pathway from manual dependency to intelligent automation across literature acquisition, sub-topic planning, and review generation. While existing frameworks have established robust technological foundations across multiple domains, they lack specific focus on social science literature review generation. Building upon these advances, this study will design an automated literature review framework tailored for social sciences, incorporating adaptive modifications to address the distinctive characteristics and requirements of social science research.

4. Proposed framework

Drawing upon the previous review of related work and established practices in social science literature reviews, we present SocLitGen, a novel framework specifically designed to automate literature review generation in the social sciences.

At its core, human literature review is fundamentally an information-seeking and knowledge acquisition process. According to the Information Search Process model (Kuhlthau, 1991), information searching encompasses at least two critical steps: formulating a focused topic and collecting information based on that focus. Schema theory further posits that for humans to acquire knowledge from information, they must comprehend, classify, and abstractly synthesize literature before forming knowledge structures (i.e., schemas) stored in memory (Rumelhart, 1980). Therefore, a complete literature review process should encompass topic planning, literature retrieval, literature comprehension, literature classification and synthesis, and overview and commentary (Bodolica & Spraggon, 2018; Fisch & Block, 2018; Paul & Criado, 2020).

This theoretical foundation, combined with established social science practices, guides our framework design with each cognitive stage mapped to corresponding computational modules: literature comprehension aligns with the literature preprocessing module, which extracts and structures key information from papers; topic planning corresponds to the topic planning module, which decomposes research questions into focused sub-topics; literature retrieval maps to the literature retrieval module, which collects relevant documents; literature classification and synthesis are realized through the review generation module and table generation module, which categorize and synthesize literature by competing theoretical perspectives in both narrative and tabular formats; overview and commentary is supported by the overview and commentary generation module for macro-level analysis; additionally, the citation formatting module ensures academic standardization. This understanding guides us to design a framework with seven corresponding modules (Fig. 2). Building upon these established stages, SocLitGen integrates advanced technologies identified in existing automated frameworks to create a comprehensive modular architecture that mirrors human cognitive processes in literature synthesis. The following sections outline the rationale and implementation details of each module.

Literature preprocessing module. This module functions to comprehend literature and present the profile of each paper through structured summaries, providing the data foundation and retrieval infrastructure for the entire framework. These summaries help scholars remember the characteristics of literature, identify differences among existing studies, and organize them more effectively (Snyder, 2019). In designing this module, the framework specifically adapts literature comprehension to meet social science requirements by ensuring that summaries contain the elements that social science scholars prioritize, such as theories, methodological approaches, samples and data, and empirical findings used to test variable relationships.

To ensure comprehensive literature coverage tailored to social science needs, this study adopts an approach inspired by MRC-Sum (Li & Xu, 2023): the system first employs a fine-tuned LLM to classify each paper into ten distinct research categories based on abstract content analysis, with categories specifically designed to capture the diversity of social science research types (detailed research

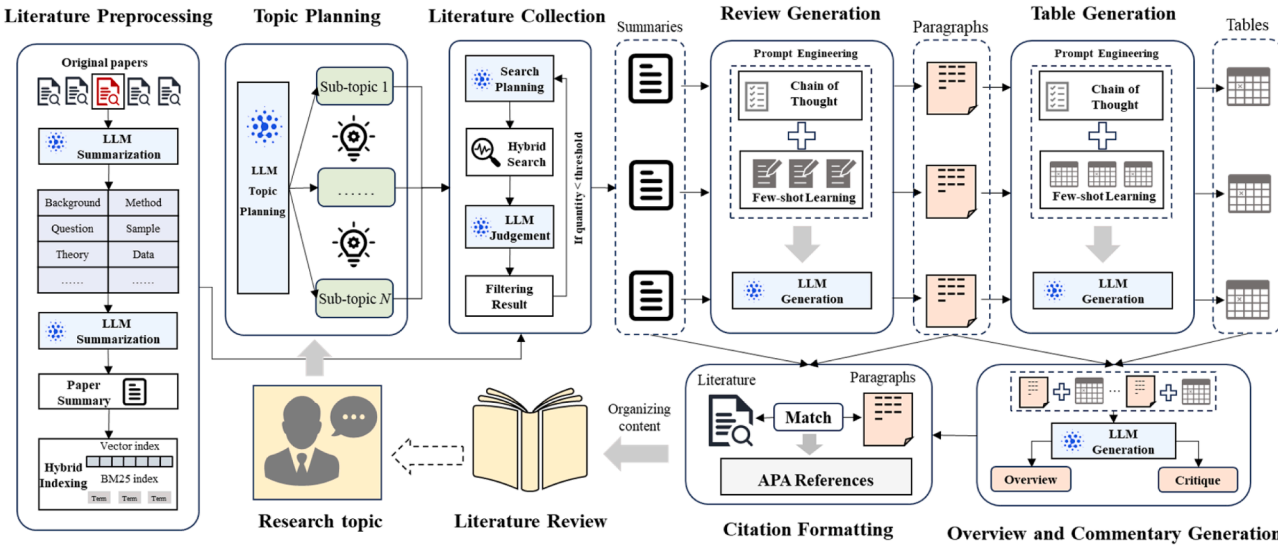


Fig. 2. Rationale and implementation of SocLitGen.

categories are presented in Table A1). Subsequently, customized question sets are designed for each paper type to extract type-specific core elements through targeted question-answering, including background, theory, hypothesis, method, sample, data, conclusion, contribution, and limitation (the definitions of extracted literature elements are provided in Table A2). Finally, LLMs integrate the extracted elements into standardized literature summaries that are both concise and comprehensive, with each summary retaining the “Author (Year)” citation for subsequent citation matching. To ensure consistency in downstream retrieval and review generation, all summaries are produced in Chinese regardless of the original language of the source paper, providing a unified linguistic foundation for subsequent processing.¹ After obtaining literature summaries, this module further constructs text inverted indexes and builds vector indexes based on a pre-trained embedding model to support efficient retrieval of social science-relevant content.

Topic planning module. This module accepts and interprets user input to output well-structured review sub-topics. We apply few-shot prompting techniques to guide a LLM in decomposing a user-specified social science research topic into a set of complementary sub-topics (typically two), each grounded in core conceptual elements (Brown et al., 2020). The importance of topic decomposition lies in providing a clear logical framework for subsequent literature collection and review writing, ensuring comprehensive coverage and in-depth exploration of research themes. This LLM-driven planning strategy has been validated across multiple prior studies for producing well-structured theoretical outlines (Liang et al., 2025; Martin-Boyle et al., 2024; Wang et al., 2020; Wang et al., 2024; Wu et al., 2025). Two criteria govern the construction of sub-topics: (1) conceptual distinctiveness—sub-topics must be non-overlapping in meaning; and (2) specificity—each sub-topic must target a concrete research dimension within its focal concept.² This approach ensures that the research topic lies at the intersection of the two sub-topics, thereby strictly meeting the requirements of social science literature reviews.

Literature collection module. Based on the retrieval database constructed by the literature preprocessing module, this module retrieves literature relevant to each sub-topic through a parallel retrieval process employing a multi-step cyclic mechanism.

The first step involves LLM-based search term planning. The system employs intelligent keyword generation technology to automatically generate diversified search term combinations for each sub-topic. This LLM-driven search term planning strategy draws from successful experiences in existing research (Agarwal et al., 2024; Sami et al., 2024; Shi et al., 2023). For example, the LitLLM framework utilizes LLMs to extract search terms from user-input abstracts (Agarwal et al., 2024), while the UR3WG framework uses LLMs to generate “pseudo-references” for user-input abstracts as search terms (Shi et al., 2023). Consistent with the summary generation in the literature preprocessing module, all search terms are generated in Chinese. This means that both Chinese- and English-language papers are treated equally during retrieval, as all summaries share a unified Chinese-language representation.

The second step implements a three-stage hybrid retrieval and screening mechanism to progressively refine document selection. First, BM25-based coarse screening is performed: the system applies the BM25 algorithm to the inverted index of literature summaries, retrieving the top 100 documents based on lexical matching scores between the Chinese search terms and Chinese summary texts. This initial filtering ensures computational efficiency while maintaining sufficient candidate coverage. Second, vector-based fine screening is conducted: the system computes cosine similarity between the query embedding (generated from search terms using the BGE-M3 model) and pre-stored document embeddings, reranking the 100 candidates and retaining the top K results (K is user-specified, typically set to 30). This semantic matching stage enhances retrieval precision by identifying documents that are conceptually relevant beyond keyword overlap. Third, LLM-based relevance judgment is applied: the system uses LLMs with Chain of Thought (CoT) prompts to perform binary classification (relevant=1, irrelevant=0) on each candidate (Wei et al., 2022). The LLM first determines whether the literature is conceptually related to the target sub-topic, then confirms whether it addresses the specific sub-category research of interest.³ Only documents classified as relevant are retained. This sequential architecture—where BM25 provides initial recall, vector similarity enhances semantic precision, and LLM judgment ensures topical alignment—achieves synergistic optimization of retrieval efficiency and quality (Beger & Henneking, 2025; Liang et al., 2025).

The third step automatically triggers iterative retrieval when LLM-validated literature for any given sub-topic falls below a pre-defined threshold (typically 15 documents per sub-topic). The system re-executes the previous steps while innovatively integrating successfully retrieved literature titles into original search terms, generating enhanced query combinations. This dynamic expansion strategy discovers additional relevant high-quality literature that initial queries may have missed, effectively ensuring the adequacy and comprehensiveness of review content (Agarwal et al., 2024). Iteration terminates when either the threshold is met or a maximum iteration limit (typically 3 rounds) is reached to prevent excessive computational costs.

Review generation module. Based on the summaries of retrieved literature, this module generates literature reviews for each sub-topic by organizing literature summaries into logically structured content. To meet social science requirements, this module organizes literature around competing theoretical perspectives. The module adopts parallelized generation strategies to simultaneously develop

¹ Standardizing all summaries into Chinese also mitigates bias in the review generation stage: LLMs exhibit uneven proficiency across languages, and generating reviews from multilingual inputs may introduce inconsistencies in synthesis quality. A unified-language input ensures that all source materials are treated equally during generation, regardless of their original language.

² Taking the research topic “the impact of digital government on green innovation” as an example, to ensure conceptual distinctiveness, this topic should first extract the two core concepts of “digital government” and “green innovation.” Subsequently, to ensure specificity, these broad concepts cannot directly serve as sub-topics. Rather, the focus must be refined to specific research dimensions within each concept, such as “research on digital government effects” and “research on influencing factors of green innovation.”

³ For example, for the “research on digital government effects” sub-topic, it verifies whether the literature involves the “digital government” concept, then further confirms whether the literature specifically elaborates on “digital government” effects rather than influencing factors or other dimensions.

content for each sub-topic, ultimately merging them into complete content (Liang et al., 2025; Wang et al., 2025). During the generation process, this module employs Chain of Thought (CoT) reasoning to systematically categorize literature under each sub-topic into multiple competing categories, systematically grouping related studies into coherent thematic groups that represent different theoretical perspectives and conceptual debates (Fisch & Block, 2018; Paul & Criado, 2020). The system first performs subdivision and classification of literature for each sub-topic, then systematically summarizes the research characteristics and core viewpoints of different categories (Brown et al., 2020; Wei et al., 2022). Throughout the generation process, LLMs are strictly required to retain “Author (Year)” identifiers from literature summaries, ensuring subsequent precise literature matching and citation generation.⁴

Table generation module. This module generates literature organization tables for each sub-topic. The module specifically designs table generation functionality, with its necessity stemming from traditional conventions in social science academic writing—providing clearer and more intuitive literature information display through tabular formats. This tabular presentation method has been widely adopted and encouraged by numerous authoritative social science journals (Paul & Criado, 2020), with top-tier journals such as *Management Review Quarterly* and *Journal of the Academy of Marketing Science* recommending the use of tables to organize and display literature information (Fisch & Block, 2018; Palmatier, 2016). This module employs LLMs to construct standardized Markdown format tables based on generated paragraph content. In the LLM prompts, this research provides detailed table generation examples and table design concepts, annotating literature in tables according to classification logic and extracting key information from each piece of literature as table columns. The generated tables not only provide researchers with effective tools for quickly grasping field development trajectories but also construct clear literature navigation maps for subsequent in-depth research.

Overview and commentary generation module. This module generates macro-level summaries and critical analyses. Given that the overview and commentary sections require macro-level summarization and critical analysis of the overall research, this module is activated after all sub-topic content is completed. Based on the global understanding capabilities of LLMs, this module first generates the overview section to systematically elaborate on the academic value of research topics and sub-topic classification logic, then generates the commentary section to deeply analyze the theoretical contributions of existing research and the potential academic value-added of the given research topic. This global overview and evaluation strategy ensures that the review possesses a complete academic framework and deep critical thinking, providing readers with comprehensive academic perspectives from whole to part, from current status to prospects.

Citation formatting module. This module performs precise matching based on “Author (Year)” identifiers retained in literature summaries using regular expressions, achieving one-to-one correspondence between main text content and references. The system automatically constructs complete reference lists by retrieving pre-stored literature metadata for matched citations and organizing them according to APA format, ensuring citation formats comply with academic standards. This precise matching mechanism not only guarantees the academic normativity of the review but also provides readers with reliable literature tracing paths, effectively avoiding omissions or errors that might occur in traditional manual organization processes.

The seven modules operate through a coordinated workflow: preprocessing transforms papers into summaries, topic planning decomposes user input, literature collection retrieves documents, review and table generation process sub-topics in parallel, overview generation provides macro-level analysis, and citation formatting ensures academic standards. Compared to traditional manual literature review processes, this modular architecture provides two key efficiency advantages. First, the literature preprocessing module front-loads the literature comprehension step by automatically extracting and standardizing core elements from papers before review writing begins. This eliminates the need for researchers to repeatedly read and analyze the same papers during synthesis. Second, the parallel processing capability allows different sub-topics to be handled simultaneously rather than sequentially, significantly reducing overall time requirements. While traditional manual approaches require researchers to complete one sub-topic before moving to the next, this framework can process multiple sub-topics concurrently, dramatically improving efficiency while maintaining the rigor and depth required for scholarly work.

Table 1 summarizes the function, input, output, and novelty of each module in the SocLitGen framework. The novelty is assessed relative to existing automated literature review frameworks across all domains, which have been predominantly developed for computer science contexts. Three modules demonstrate significant innovations: the Literature Preprocessing Module introduces element extraction-based summary generation tailored to social science requirements, extracting theories, methodological approaches, samples, and empirical findings rather than task definitions and technical solutions; the Review Generation Module employs classification-based synthesis that organizes literature around competing theoretical perspectives rather than technological evolution patterns; and the Table Generation Module represents an entirely new contribution, as no previous framework in any domain has systematically integrated literature organization tables into the automated generation pipeline. Other modules adapt established approaches from computer science frameworks to address social science requirements. Appendix B provides a complete case study demonstrating the outputs of all modules except the literature preprocessing module, illustrating how SocLitGen processes a research topic through its entire pipeline from topic planning to citation formatting.

⁴ For generation parameters, we adopt each LLM provider’s default temperature setting (typically a relatively high value to encourage diverse expressions and avoid repetitive outputs) and set max_tokens to 8000 to accommodate comprehensive synthesis. Detailed prompt templates, few-shot examples, and complete implementation code are publicly available in our GitHub repository to ensure experimental reproducibility and transparency.

Table 1
Module functions and novelty in the SocLitGen framework.

Module	Function	Input	Output	Novelty
Literature Preprocessing Module	Generate structured summaries with social science elements	Raw papers	Structured literature summaries	Element extraction-based summary generation
Topic Planning Module	Decompose research topic into sub-topics	User research topic	Conceptually distinct sub-topics	Few-shot planning
Literature Collection Module	Retrieve relevant literature for each sub-topic	Sub-topics	Literature metadata and summaries	Three-step iterative retrieval
Review Generation Module	Organize summaries into review content	Literature summaries	Review content by competing categories	Classification-based synthesis
Table Generation Module	Generate literature organization tables	Review content	Markdown tables	Entirely new module
Overview and Commentary Generation Module	Generate macro-level overview and commentary	All sub-topic contents	Overview and commentary sections	None
Citation Formatting Module	Format citations according to academic standards	Sub-topic contents and literature metadata	APA format citation list	None

5. Experiments and results

To address the two research questions formulated in Section 2, this study constructs a large-scale social science literature database, develops a comprehensive evaluation framework specifically designed for assessing social science literature review quality, and selects representative baselines including human writing and existing automated frameworks from other domains. Baseline comparisons are then conducted to answer RQ1, while cross-context evaluations across three dimensions—disciplinary domain, underlying LLM, and literature scale—are conducted to answer RQ2. The following subsections detail the dataset, evaluation metrics, baselines, and experimental results. To rigorously evaluate the effectiveness of the proposed SocLitGen framework, we constructed a custom large-scale dataset rather than relying on existing public databases. This decision stems from two key limitations of public databases: first, they primarily source papers from preprint repositories such as arXiv, whereas high-quality social science research is predominantly published in peer-reviewed journals; second, they lack comprehensive coverage of Chinese-language scholarship, which constitutes a substantial portion of social science research output.

Our custom dataset spans 26 core disciplines—including management, economics, education, psychology, and sociology—and comprises literature from 1432 authoritative academic journals, with data collection extending through August 20, 2025. These include both CSSCI-indexed Chinese journals and SSCI-indexed international journals. Journal selection followed systematic criteria to ensure disciplinary representativeness and quality consistency. For English-language literature, we referenced the Web of Science (WoS) category classification and selected journals ranked in JCR Q1-Q2 quartiles within each discipline. For Chinese-language literature, we included all CSSCI-indexed journals within each corresponding discipline.

To ensure comprehensive coverage while minimizing bias and duplication, we implemented a systematic two-stage collection process. First, we exported complete publication lists of selected journals from Web of Science (for English sources) and CNKI (for Chinese sources), obtaining structured metadata including DOI and article links. Second, using these standardized identifiers, we developed automated web crawlers to retrieve PDFs through institutional access to databases such as ScienceDirect and JSTOR for English literature, and CNKI, Wanfang, and VIP for Chinese literature. This export-first approach ensures exhaustive journal coverage without sampling bias and effectively prevents duplicate entries. Following PDF retrieval and a rigorous quality-control process that removed corrupted files and non-research articles (editorials, book reviews, conference announcements), the final dataset included 1559,141 high-quality papers, comprising 183,886 Chinese-language papers (11.8 %) and 1375,255 English-language papers (88.2 %) across 1432 journals (294 Chinese-language and 1138 English-language journals spanning 26 disciplines).⁵ For each entry, we extracted standard metadata—title, authors, journal, volume, issue, publication year, abstract, and keywords—and parsed full texts using PDF-to-text conversion tools. The disciplinary distribution of journals and papers is shown in Fig. 3.

To facilitate document retrieval, both inverted indexes and vector indexes were constructed for all entries based on their structured summaries. Specifically, the inverted index was built by tokenizing summaries and removing stop words, while the vector index was generated using the BGE-M3 pre-trained model (BAAI) to produce 1024-dimensional cross-lingual summary embeddings (Chen et al., 2024). The dataset was deployed on a high-performance Elasticsearch cluster to support efficient querying and retrieval across all stages of the SocLitGen framework. The database field descriptions are shown in Table A3.

5.1. Evaluation metrics

To ensure objective, consistent, and reliable evaluation, this study recruited five PhD candidates from a Ministry of Education-affiliated university in central China as expert evaluators. All evaluators underwent a comprehensive training session on the evaluation framework and scoring criteria, followed by practice exercises on sample literature reviews to ensure consistency in under-

⁵ The complete list of included journals is available in our GitHub repository. A detailed breakdown of journal and paper counts by discipline and language is provided in Appendix A (Figs. A1 and A2).

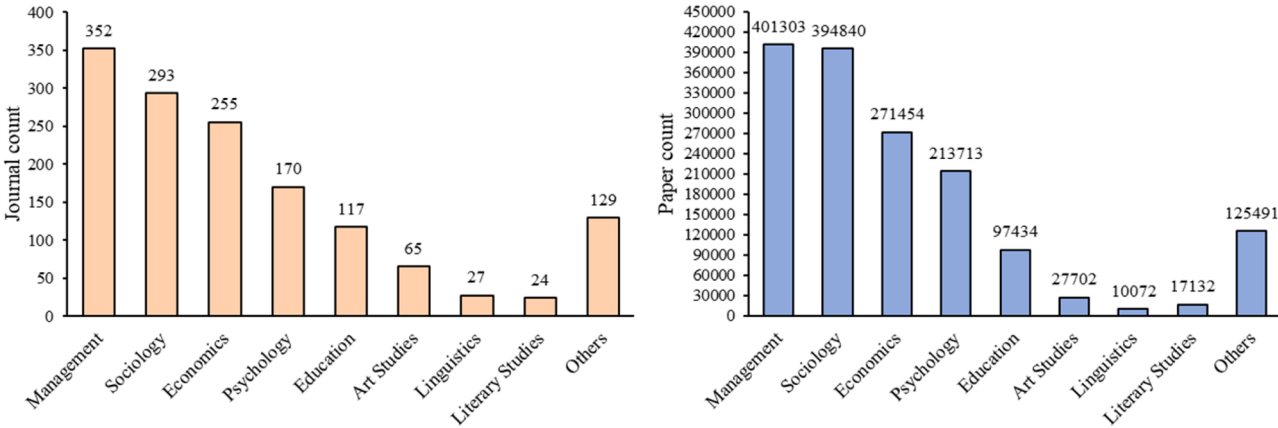


Fig. 3. Disciplinary distribution of social science literature dataset.

standing and application of the evaluation metrics. Pre-annotation reliability tests were conducted using Intraclass Correlation Coefficients (ICC) for content quality ratings and Fleiss' Kappa (κ) for citation relevance judgments.⁶ Results demonstrated high inter-rater consistency ($\text{ICC} > 0.75$ and $\kappa > 0.70$), confirming the evaluators' readiness for the formal assessment task. Each evaluator was then required to independently assess all generated literature reviews across both citation quality and content quality dimensions. This study adopts an evaluation framework inspired by prior research, assessing the effectiveness of the SocLitGen framework from two core perspectives: citation quality and content quality, in order to ensure comprehensive and comparable results.

5.1.1. Citation quality metrics

Citation quality evaluation is a key component in assessing the academic credibility of automatically generated literature reviews. It primarily examines the coverage and relevance of citations within the generated text. In existing studies, scholars commonly use citation recall and citation precision as the core quantitative metrics for evaluating citation quality (Asai et al., 2024; Wang et al., 2025; Wang et al., 2024). In line with these studies, this study adopts both citation recall and citation precision to assess the citation performance of the SocLitGen framework.

For the human evaluation implementation, each evaluator was provided with both the generated literature review and the input literature set used for generation. The evaluators were tasked with manually identifying and counting specific parameters to calculate the citation quality metrics:

Citation Recall. Evaluators examined the generated review to identify all cited documents and determined whether each document from the input literature set $D = d_1, d_2, \dots, d_n$ was actually referenced in the review. For calculation purposes, evaluators counted two key parameters: (1) the total number of documents in the input literature set (n), and (2) the number of input documents actually cited in the review. Citation recall follows the definition proposed by Wang et al. (2025), referring to the degree of coverage of the input candidate literature in the generated review. This metric reflects the efficiency with which available literature resources are utilized in the generation process. Formally, for an input document set $D = d_1, d_2, \dots, d_n$, and the subset actually cited $C_{\text{cited}} \subseteq D$, The indicator function $I(d_i \in C_{\text{cited}})$ is defined to denote whether document d_i has been cited. The citation recall calculation formula is:

$$\text{Citation recall} = \frac{\sum_{i=1}^n I(d_i \in C_{\text{cited}})}{n}$$

Citation Precision. For each document cited in the generated review, evaluators assessed its topical relevance to the target research topic using their expert judgment. Evaluators identified and counted two essential parameters: (1) the total number of documents cited in the generated review (m), and (2) the number of cited documents that were topically relevant to the research topic. Citation precision follows the definition provided by Asai et al. (2024), referring to the topical relevance between the cited documents and the target research topic. This metric reflects the accuracy and restraint in citation selection, helping to avoid excessive or irrelevant referencing. Let the target research topic be denoted as T , and the set of cited documents in the generated review be $C_{\text{cited}} = c_1, c_2, \dots, c_m$. A binary relevance function $R(c_j, T) \in \{0, 1\}$ is defined to indicate whether document c_j is relevant to topic T . The citation precision is computed as:

$$\text{Citation precision} = \frac{\sum_{j=1}^m R(c_j, T)}{m}$$

Each evaluator independently determined the relevance of cited documents based on their expertise in the respective social science domains, and the final scores represented the average across all five evaluators.

5.1.2. Content quality metrics

In the evaluation system for automatic literature review generation systems, content-level quality assessment typically encompasses multiple core dimensions. Commonly adopted evaluation metrics in academia include topic relevance (assessing the alignment between generated content and research topics) (Asai et al., 2024; Chen et al., 2022; Li et al., 2022; Shi et al., 2023; Wang et al., 2025), structural rationality (examining the logic and hierarchy of content organization) (Chen et al., 2022; Li et al., 2022; Palani et al., 2023; Shi et al., 2023; Wang et al., 2025; Wang et al., 2024), language fluency (measuring the naturalness and readability of textual expression) (Li et al., 2022; Shi et al., 2023), language conciseness (evaluating the refinement degree and information density of expression) (Chen et al., 2022), and academic professionalism (examining overall professionalism and academic normativity) (Sami et al., 2024). These metrics comprehensively reflect the overall performance of automatically generated reviews in terms of content quality, expressive effectiveness, and academic value from different perspectives.

Considering evaluation practicality and independence between metrics, we ultimately selected three core evaluation dimensions: academic professionalism, structural rationality, and language fluency. The main reasons for choosing these three metrics while

⁶ Fleiss' Kappa was adopted for citation relevance judgments as it is specifically designed for scenarios involving more than two raters. For each cited document, each of the five evaluators independently assigned a binary relevance label (relevant = 1, irrelevant = 0). Fleiss' Kappa is computed as $\hat{\kappa} = (B - W) / [B + (m - 1)W]$, where $B = \frac{1}{n} \sum_i (x_i - m_i p)^2 / m_i$ is the between-subjects mean square, $W = \frac{1}{n(m-1)} \sum_i x_i (m_i - x_i) / m_i$ is the within-subjects mean square, n denotes the number of cited documents under evaluation, x_i is the number of raters who judged document i as relevant, $m_i = 5$ is the number of raters, and $p = \sum_i x_i / (nm)$ is the overall proportion of positive ratings.

excluding the other two are: first, topic relevance overlaps conceptually with literature relevance evaluation in the literature retrieval stage, potentially causing evaluation redundancy; second, language conciseness easily confuses with text length, making it difficult to accurately reflect the essential characteristics of content quality. In contrast, academic professionalism evaluates the domain appropriateness of generated content from a global perspective, structural rationality examines content logical coherence from an organizational framework level, and language fluency measures text naturalness from a sentence expression level. These three dimensions are mutually independent and can comprehensively cover key elements of review quality. Specific items for the three metrics are shown in Table A4.

For the human evaluation implementation, each of the five evaluators independently scored all generated literature reviews on a 5-point Likert scale across the three content quality dimensions. Detailed scoring rubrics were provided for each dimension (see Table A4), and evaluators were instructed to consider domain-specific requirements for social science literature reviews. All individual scores from the 25 topics $\times$ 5 evaluators were included in the statistical analysis to ensure robust and reliable assessment results.

5.2. Baselines

To comprehensively evaluate the SocLitGen framework, this study designed a systematic evaluation involving 25 literature review generation tasks across social science research topics (13 in management and 12 in economics). These topics were proposed by graduate students to facilitate cross-disciplinary performance comparison. The selection of economics and management was strategic, given their contrasting knowledge utilization patterns: economics serves as a net knowledge exporter while management functions primarily as a knowledge importer in academic literature (Lockett & McWilliams, 2005). This disciplinary contrast creates extreme testing conditions that effectively reveal the framework's performance boundaries and generalizability limits. This study constructs three comparison baselines: first, AutoSurvey (Wang et al., 2024), an authoritative framework for literature review generation in computer science; second, Graduate Student Baseline, which involves graduate student completion of literature review tasks; and third, Naive RAG Baseline, which adopts the standard Retrieval-Augmented Generation paradigm. The latter two baselines follow the evaluation paradigm established by AutoSurvey. All literature reviews in this evaluation were generated in Chinese, the native language of the graduate student participants, ensuring that language proficiency does not confound the quality assessment.

Graduate Student Baseline. This baseline involves master's students completing the literature review generation task under controlled conditions. The process consists of three steps: first, 25 master's students from management and economics programs at a Ministry of Education-affiliated university in central China proposed research topics based on their academic interests; second, these students conducted literature retrieval exclusively within the unified literature dataset constructed for this study, ensuring experimental control and avoiding interference from external knowledge sources; third, they independently wrote complete literature reviews within a 90-min session (approximately two class periods), with no structural format requirements imposed so as to allow authentic expression of individual scholarly competence, and with citations formatted in APA style traceable to the unified dataset.⁷

Naive RAG Baseline. This baseline adopts the standard Retrieval-Augmented Generation (RAG) paradigm, implementing review generation through a three-stage process. First, based on user-input research topics, it calls the hybrid retrieval module to perform literature screening combining lexical tokenization and semantic matching in the constructed knowledge base; second, it integrates retrieved literature metadata (title, author, publication year) and abstract information into structured prompt templates for input to LLMs; finally, through carefully designed instruction prompts, it guides the model to generate standardized Chinese academic reviews including research status review, method comparative analysis, problem identification, and future development directions. RAG approaches have been extensively adopted by numerous frameworks for review generation tasks (Agarwal et al., 2024; Asai et al., 2024; Li et al., 2024). This baseline setup follows the convention of mainstream research such as AutoSurvey in using RAG methods as core comparison baselines, ensuring the scientific validity and comparability of the evaluation framework.

AutoSurvey Baseline. Beyond the conventional baselines above, we additionally implement AutoSurvey as a third baseline to represent a fundamentally different workflow approach (Wang et al., 2024). Since the original AutoSurvey framework was designed for English-language computer science literature, we reconstructed its core methodology to operate on our Chinese-English social science literature database while preserving its essential workflow logic. The key distinction lies in the sequencing of topic planning and literature retrieval: AutoSurvey first conducts literature retrieval, then generates the review outline based on retrieved literature content, and finally performs parallel section generation. This retrieval-first approach contrasts with SocLitGen's planning-first strategy, where topic decomposition precedes targeted literature collection. The AutoSurvey methodology has also been adopted as a baseline in subsequent work (Wang et al., 2025), validating its representativeness as an alternative automated framework.

To ensure experimental consistency and comparability, the SocLitGen framework, Naive RAG baseline, and AutoSurvey baseline uniformly adopt the following configuration: using review topics proposed by graduate students as input and setting the literature count to 15 per sub-topic (30 in total). All three automated methods employ the GLM-4 LLM as the driving model (Team GLM, 2024). All four methods (including graduate student baseline) undergo content quality dimension evaluation, while citation quality evaluation only applies to the three automated baselines, as graduate students with professional backgrounds approach ideal levels in citation selection, making comparability with automated systems relatively low.

⁷ The use of AI-assisted writing tools was strictly prohibited throughout the session, and compliance was enforced through in-person supervision.

5.3. Main results

This section presents the experimental results obtained under a standardized system configuration to ensure reproducibility and transparency. The backend infrastructure consisted of two primary components: (1) Elasticsearch 8.11.3 deployed on a Linux cloud server with 64GB RAM and 32 CPU cores for storing and indexing the literature dataset, and (2) MongoDB 7.0.12-jammy deployed on the same server for recording generation task results and maintaining task logs. The testing environment ran on Windows 11 with Python 3.11, equipped with a workstation featuring 16 CPU cores and 32GB RAM. Under this configuration, the complete system pipeline—from topic input to final review generation—required an average of 47.2 s per task, demonstrating the framework's computational efficiency and practical applicability for real-world scholarly workflows. The following subsections report performance comparisons and analysis results based on this experimental setup.⁸

5.3.1. Comparison with baselines

This study conducted paired-samples *t*-tests to evaluate SocLitGen's performance against three baselines ($n = 125$ evaluations: 25 topics $\times$ 5 evaluators) across citation quality and content quality dimensions.

Table 2 presents the citation quality evaluation results. SocLitGen significantly outperformed Naive RAG on Citation Recall, $t(124)=14.40, p < 0.001$, and Citation Precision, $t(124)=15.03, p < 0.001$. However, SocLitGen showed no significant difference from AutoSurvey on Citation Recall, $t(124)=1.10, p = 0.275$, or Citation Precision, $t(124)=0.91, p = 0.363$.

Table 3 presents the content quality evaluation results. On Academic Professionalism, SocLitGen significantly outperformed Naive RAG, $t(124)=12.66, p < 0.001$, AutoSurvey, $t(124)=4.52, p < 0.001$, and Graduate Students, $t(124)=3.83, p < 0.001$. Graduate Students also significantly exceeded Naive RAG, $t(124)=-10.10, p < 0.001$. On Structural Rationality, SocLitGen significantly surpassed Naive RAG, $t(124)=8.25, p < 0.001$, AutoSurvey, $t(124)=7.57, p < 0.001$, and Graduate Students, $t(124)=4.58, p < 0.001$. Graduate Students also significantly outperformed Naive RAG, $t(124)=-4.44, p < 0.001$. On Language Fluency, SocLitGen showed no significant difference from Naive RAG, $t(124)=-0.37, p = 0.713$, but significantly exceeded AutoSurvey, $t(124)=7.85, p < 0.001$, and Graduate Students, $t(124)=6.90, p < 0.001$.

These results demonstrate that SocLitGen achieves significant advantages over Naive RAG in citation quality, which can be attributed to the framework's three-step iterative retrieval mechanism that effectively identifies relevant literature beyond simple retrieval pipelines. For content quality, SocLitGen outperforms all baselines in Academic Professionalism and Structural Rationality, which is likely due to the element extraction and classification-based synthesis modules that effectively capture social science conventions. Additionally, SocLitGen demonstrates superior Language Fluency compared to AutoSurvey and Graduate Students, suggesting that the planning-first workflow produces more coherent narratives than retrieval-first approaches or graduate student writing.

5.3.2. Cross-disciplinary performance comparison

To examine the generalizability of SocLitGen across different social science domains, we conducted independent-samples *t*-tests to compare performance between management topics (13 topics $\times$ 5 evaluators = 65 evaluations) and economics topics (12 topics $\times$ 5 evaluators = 60 evaluations), two representative fields within social sciences.

Table 4 presents the comparative results across all evaluation metrics. For citation quality, Citation Recall showed no significant difference, $t(123)=0.03, p = 0.975$, nor did Citation Precision, $t(123)=0.84, p = 0.402$. For content quality, Academic Professionalism, $t(123)=0.66, p = 0.509$, Structural Rationality, $t(123)=0.80, p = 0.427$, and Language Fluency, $t(123)=0.71, p = 0.480$, all demonstrated no significant differences between disciplines.

These results demonstrate that SocLitGen maintains consistent performance across management and economics disciplines. This cross-disciplinary robustness suggests that the framework's core mechanisms—which capture field-appropriate information and organize literature around competing perspectives—are sufficiently general to accommodate different theoretical traditions within these two social science disciplines.

5.3.3. Cross-model performance comparison

To further clarify the performance of the SocLitGen framework under different task contexts, this study replaced the text generation model in the framework and re-evaluated the multi-dimensional metrics of content quality dimensions. Specifically, beyond the original GLM-4 model, this study additionally selected three LLMs: GPT-4o, DeepSeek-V3, and Qwen2.5 (DeepSeek-AI, 2024; OpenAI, 2023; Qwen Team, 2024). These three models ranked at the top in text comprehension and generation ability scores in the March 2025 SuperCLUE evaluation⁹: GPT-4o scored first among closed-source general models, while DeepSeek-V3 and Qwen2.5 scored first and second among open-source models, respectively. Based on these three new models, this study regenerated literature reviews for the 25 topics and evaluated them together with the literature reviews generated by the original GLM-4 model. The comparative results are presented in Table 5. According to the multi-model performance comparison results of the SocLitGen framework, significant differences were observed across models and evaluation dimensions.

On Academic Professionalism, significant differences emerged across all four models. Qwen2.5 achieved the highest performance,

⁸ We acknowledge the non-deterministic nature of LLM outputs with non-zero temperature. Each review was generated once, and all reported results reflect single-run performance.

⁹ A benchmark released by the CLUE community, for evaluating Chinese-accessible LLMs. Conducting regular multi-dimensional assessments of representative domestic and international models and publishes corresponding leaderboards (L. Xu et al., 2023).

Table 2
Citation quality evaluation.

Metrics	SocLitGen		Naive RAG		AutoSurvey	
	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI
Citation Recall	0.565	[0.546, 0.585]	0.404	[0.395, 0.412]	0.561	[0.538, 0.583]
Citation Precision	0.614	[0.585, 0.642]	0.361	[0.346, 0.376]	0.605	[0.571, 0.639]

Table 3
Content quality evaluation.

Metrics	SocLitGen		Naive RAG		AutoSurvey		Graduate Students	
	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI
Academic Professionalism	3.520	[3.319, 3.721]	1.968	[1.814, 2.122]	3.112	[2.872, 3.352]	3.048	[2.904, 3.192]
Structural Rationality	4.504	[4.415, 4.593]	3.568	[3.368, 3.768]	3.800	[3.604, 3.996]	4.112	[3.962, 4.262]
Language Fluency	4.272	[4.124, 4.420]	4.312	[4.167, 4.457]	3.600	[3.373, 3.827]	3.408	[3.198, 3.618]

Table 4
Cross-disciplinary performance comparison.

Metrics	Management		Economics	
	Mean	95 % CI	Mean	95 % CI
Citation Recall	0.566	[0.538, 0.593]	0.565	[0.536, 0.594]
Citation Precision	0.625	[0.587, 0.664]	0.601	[0.557, 0.645]
Academic Professionalism	3.585	[3.301, 3.868]	3.450	[3.159, 3.741]
Structural Rationality	4.538	[4.414, 4.663]	4.467	[4.337, 4.597]
Language Fluency	4.323	[4.122, 4.524]	4.217	[3.993, 4.440]

significantly outperforming DeepSeek-V3, $t(124)=-2.62$, $p = 0.010$, as well as GLM-4, $t(124)=-6.38$, $p < 0.001$, and GPT-4o, $t(124)=-11.38$, $p < 0.001$. Among the remaining models, DeepSeek-V3 demonstrated superior performance compared to both GLM-4, $t(124)=-4.37$, $p < 0.001$, and GPT-4o, $t(124)=-8.94$, $p < 0.001$. GLM-4 also significantly outperformed GPT-4o, $t(124)=3.41$, $p < 0.001$, establishing a clear performance hierarchy: Qwen2.5 > DeepSeek-V3 > GLM-4 > GPT-4o.

On Structural Rationality, a similar pattern emerged. Qwen2.5 maintained its dominant position, significantly surpassing GLM-4, $t(124)=-4.34$, $p < 0.001$, DeepSeek-V3, $t(124)=-6.44$, $p < 0.001$, and GPT-4o, $t(124)=-12.01$, $p < 0.001$. GLM-4 ranked second, significantly exceeding both DeepSeek-V3, $t(124)=3.48$, $p < 0.001$, and GPT-4o, $t(124)=9.13$, $p < 0.001$. DeepSeek-V3 also significantly outperformed GPT-4o, $t(124)=-5.69$, $p < 0.001$, maintaining the same performance hierarchy observed in Academic Professionalism: Qwen2.5 > GLM-4 > DeepSeek-V3 > GPT-4o.

On Language Fluency, however, the performance pattern reversed. GLM-4 emerged as the top performer, significantly exceeding Qwen2.5, $t(124)=2.20$, $p = 0.030$, DeepSeek-V3, $t(124)=5.68$, $p < 0.001$, and GPT-4o, $t(124)=5.67$, $p < 0.001$. Qwen2.5 secured the second position, significantly outperforming both DeepSeek-V3, $t(124)=-4.20$, $p < 0.001$, and GPT-4o, $t(124)=-4.52$, $p < 0.001$. DeepSeek-V3 and GPT-4o showed comparable performance with no significant difference, $t(124)=-0.35$, $p = 0.723$, establishing a new hierarchy: GLM-4 > Qwen2.5 > DeepSeek-V3 $\approx$ GPT-4o.

These results demonstrate that Qwen2.5 achieves superior performance in Academic Professionalism and Structural Rationality across all other models, suggesting that this model possesses stronger capabilities in adhering to social science disciplinary conventions and constructing logically coherent arguments. Additionally, GLM-4 demonstrates advantages in Language Fluency over all other models, which may reflect its optimization for natural language generation in Chinese contexts.

5.3.4. Cross-literature scale performance comparison

To evaluate the impact of literature input size on SocLitGen performance, this study compared results using different retrieval thresholds per sub-topic: small-scale (15 papers per sub-topic, totaling 30 papers across two sub-topics) and large-scale (30 papers per sub-topic, totaling 60 papers across two sub-topics). The analysis revealed differentiated performance patterns across content quality dimensions, as detailed in Table 6.

On Academic Professionalism, the large-scale condition significantly outperformed the small-scale condition, $t(124)=4.78$, $p < 0.001$. Similarly, on Structural Rationality, large-scale literature significantly exceeded small-scale literature, $t(124)=3.57$, $p < 0.001$. However, on Language Fluency, the pattern reversed: small-scale literature significantly outperformed large-scale literature, $t(124)=-5.99$, $p < 0.001$.

These results indicate that large-scale literature sets yield significant advantages in Academic Professionalism and Structural Rationality, attributable to richer empirical foundations that enable more nuanced theoretical perspectives and comprehensive coverage of competing viewpoints. However, small-scale literature sets demonstrate superior Language Fluency, as processing fewer papers reduces organizational complexity and allows LLMs to maintain smoother narrative flow. Post-hoc examination of reviews with

Table 5
Cross-model performance comparison.

Metrics	GLM-4		GPT-4o		DeepSeek-V3		Qwen2.5	
	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI	Mean	95 % CI
Academic Professionalism	3.520	[3.319, 3.721]	3.112	[2.966, 3.258]	4.072	[3.935, 4.209]	4.320	[4.177, 4.463]
Structural Rationality	4.504	[4.415, 4.593]	3.512	[3.313, 3.711]	4.192	[4.045, 4.339]	4.760	[4.684, 4.836]
Language Fluency	4.272	[4.124, 4.420]	3.504	[3.305, 3.703]	3.552	[3.357, 3.747]	4.056	[3.919, 4.193]

lower fluency scores suggests that this trade-off is primarily driven by retrieval noise: as input size increases, documents with weaker topical relevance that pass through the screening stage introduce tangential content that disrupts narrative coherence. This issue may naturally diminish as the literature database expands, since a larger corpus would yield a higher proportion of genuinely relevant documents within the retrieved set, thereby reducing retrieval noise at larger input scales.

5.4. Sensitivity analysis of the retrieval corpus

To examine the robustness of SocLitGen's performance across different data sources, this study conducted a sensitivity analysis by implementing an alternative framework configuration. In this modified setup, the literature collection module was reconfigured to utilize open domain retrieval through the Semantic Scholar API's native capabilities, which were then augmented with the LLM-based screening mechanism originally proposed in the framework. Specifically, the open domain retrieval system first automatically generated search terms for each sub-topic, then iteratively executed API-based retrieval combined with LLM fine screening until the target number of documents was collected for each sub-topic. Critically, the literature preprocessing module was repositioned to occur after the retrieval stage rather than before it, enabling real-time extraction and summarization of papers with available PDF documents. All other modules—including topic planning, review generation, table generation, overview and commentary generation, and citation formatting—remained unchanged to ensure controlled comparison conditions. This configuration provided a different data foundation while maintaining the core architectural integrity of the SocLitGen framework.

The sensitivity analysis revealed nuanced performance variations across different metrics, as shown in Table 7. For content quality dimensions, Academic Professionalism showed no significant difference between sources, $t(124) = -0.65$, $p = 0.519$, nor did Language Fluency, $t(124) = 0.24$, $p = 0.813$, or Structural Rationality, $t(124) = 0.24$, $p = 0.812$. However, for citation quality metrics, the alternative data source demonstrated significantly superior performance on both Citation Recall, $t(124) = -2.75$, $p = 0.007$, and Citation Precision, $t(124) = -3.59$, $p < 0.001$.

These results demonstrate that the alternative data source achieves significant advantages in citation quality, which can be attributed to its real-time retrieval approach accessing a broader literature base and Semantic Scholar's citation network-based ranking algorithms. The consistent content quality across both data sources confirms that SocLitGen's core architectural design maintains robustness regardless of corpus source.

6. Discussions

This study formulated two research questions to guide the investigation. RQ1 examines how the proposed framework performs in social science literature review generation compared to existing approaches, addressing the gap that existing frameworks primarily target computer science contexts and struggle to adapt to social science requirements, particularly in literature comprehension and review organization. RQ2 examines how the framework's performance varies across different task contexts, recognizing that understanding performance boundaries across different models, literature scales, and disciplinary contexts is essential for practical deployment.

To address RQ1, we constructed a large-scale literature database covering 1559,141 papers, then tested the SocLitGen framework and compared its performance against three baselines: Naive RAG, AutoSurvey, and Graduate Students. Baseline comparisons demonstrate the value of domain-specific design. In citation quality, SocLitGen outperforms Naive RAG by 39.9 % in recall and 70.1 % in precision, while achieving comparable performance to AutoSurvey. In content quality, SocLitGen demonstrates consistent advantages: SocLitGen surpasses AutoSurvey by 13.1 %, 18.5 %, and 18.7 % in academic professionalism, structural rationality, and language fluency, respectively; surpasses graduate student writing by 15.5 %, 9.5 %, and 25.4 % in the same dimensions. The substantial performance gaps between SocLitGen and both generic (Naive RAG) and alternative automated approaches (AutoSurvey) demonstrate that adapting to social science requirements yields tangible benefits. Collectively, these baseline comparisons confirm the necessity of

Table 6
Cross-scale performance comparison.

Metrics	Small literature set		Large literature set	
	Mean	95 % CI	Mean	95 % CI
Academic Professionalism	3.520	[3.319, 3.721]	4.120	[3.966, 4.274]
Structural Rationality	4.504	[4.415, 4.593]	4.712	[4.631, 4.793]
Language Fluency	4.272	[4.124, 4.420]	3.544	[3.357, 3.731]

Table 7
Sensitivity analysis of retrieval corpus.

Metrics	Original Source		Alternative Source	
	Mean	95 % CI	Mean	95 % CI
Academic Professionalism	3.520	[3.319, 3.721]	3.616	[3.409, 3.823]
Structural Rationality	4.504	[4.415, 4.593]	4.488	[4.399, 4.577]
Language Fluency	4.272	[4.124, 4.420]	4.248	[4.094, 4.402]
Citation Recall	0.565	[0.546, 0.585]	0.603	[0.585, 0.622]
Citation Precision	0.614	[0.585, 0.642]	0.684	[0.658, 0.711]

developing specialized frameworks for social science literature review generation rather than relying on generic or repurposed systems.

To address RQ2, we conducted systematic evaluations across three task contexts, revealing both capabilities and boundaries of the framework. First, cross-disciplinary analysis reveals consistent performance across management and economics, despite their contrasting positions as knowledge importer and exporter, providing initial evidence of cross-field generalizability. This suggests that the framework's core mechanisms are robust to accommodate different theoretical traditions within social sciences. Second, cross-model evaluation demonstrates that model selection critically shapes output quality in ways that diverge from general-purpose performance rankings: Qwen2.5 excels at academic rigor and structural organization, while GLM-4 produces more fluid prose. Most striking is GPT-4o's relatively weak showing across all dimensions, despite its documented superiority in multilingual translation, mathematical reasoning, and general knowledge tasks (Feng et al., 2024; Li et al., 2025; Liu et al., 2024). This implies that general-purpose benchmark reputation does not reliably predict performance in domain-specific academic writing tasks. Third, literature scale analysis uncovers an inherent tension between comprehensiveness and readability: expanding from small-scale (15 papers per sub-topic, 30 in total) to large-scale (30 papers per sub-topic, 60 in total) enhances academic professionalism and structural rationality, yet at the cost of reduced linguistic fluency. This indicates that literature scale constitutes a meaningful performance boundary of the framework, likely linked to the inherent constraints of the self-built literature database.

7. Implications

This study proposes SocLitGen, an automated literature review generation framework specifically designed for social sciences. Through systematic evaluation, the framework demonstrates superior performance compared to existing approaches and exhibits robust, context-dependent characteristics across different operational conditions. These findings offer important theoretical and practical implications for both academic research and technology development.

7.1. Theoretical implications

Based on the above research findings, this study presents three critical theoretical contributions to the existing research on automated literature review generation.

The first theoretical implication is that this study demonstrates the effectiveness of expanding automated literature review technology across disciplinary boundaries. Previous studies have predominantly concentrated on computer science, chemistry, and industrial manufacturing domains, essentially limiting automated review generation research to natural science and engineering fields while leaving social sciences unexplored. In contrast, our SocLitGen framework addresses the unique characteristics of social science literature, demonstrating that automated review technology can be successfully adapted to humanities and social sciences through domain-specific design considerations. This disciplinary expansion reveals that automated literature review systems can transcend their original technical domains when properly tailored to different epistemological traditions and organizational conventions.

The second theoretical implication of our study is that SocLitGen highlights the necessity of incorporating comprehensive workflow refinements across multiple dimensions to enhance automated review quality. Unlike previous works that adopt relatively simplified generation processes, our modular architecture introduces several key innovations. The element extraction-based summary generation in the literature preprocessing module transforms papers into structured summaries specifically designed for social science requirements, achieving a "quality in, quality out" principle. The few-shot planning approach in the topic planning module enables more nuanced topic decomposition that aligns with social science research paradigms and theoretical frameworks. The three-step iterative retrieval process ensures comprehensive literature collection through systematic filtering and refinement. The classification-based synthesis approach in the review generation module organizes literature by competing categories rather than simple chronological or thematic groupings. Additionally, the entirely new table generation module provides structured literature organization that enhances readability and comprehensiveness. These innovations demonstrate that automated literature review systems require domain-specific adaptations and modular design principles to achieve high-quality academic output tailored to the epistemological traditions of social sciences.

The third theoretical implication of this study is that our systematic evaluation framework reveals the critical importance of understanding task context dependencies in automated literature review performance. Previous studies have relied on static evaluation paradigms that compare proposed methods with baselines under fixed experimental conditions, but this compromises the understanding of framework adaptability and performance boundaries. In contrast, our cross-model, cross-literature scale, and cross-

disciplinary experiments demonstrate that different LLMs exhibit differentiated performance characteristics when driving the same framework, that literature scale impacts various quality dimensions in complex, non-uniform ways, and that performance remains consistent across different social science disciplines. These findings provide essential theoretical insights into the task context adaptability mechanisms of AI systems, enabling more informed deployment decisions and performance optimization strategies.

7.2. Practical implications

The findings of this study offer practical insights for three key stakeholder groups: social science researchers, developers of automated literature review technology, and technology standardization bodies.

Social science researchers can consider incorporating automated literature review tools like SocLitGen into auxiliary stages of academic research and flexibly select configuration parameters of the SocLitGen framework according to specific research needs. When research tasks emphasize academic rigor and logical structure, Qwen2.5 should be prioritized as the driving model, with larger-scale literature sets provided as input to achieve higher academic professionalism and structural rationality; when research tasks emphasize expression fluency and readability, GLM-4 should be selected as the driving model, with appropriate control of input literature scale to balance content quality and language expression effects. The framework's consistent performance across different social science disciplines suggests that these configuration strategies can be applied broadly within the social sciences without field-specific adjustments. This differentiated configuration strategy can help researchers maximize the application effectiveness of the framework under different research contexts. However, it is important to emphasize that SocLitGen functions as an assistive tool rather than a replacement for human scholarly judgment. Researchers should critically evaluate and refine automatically generated reviews, incorporating their domain expertise, theoretical understanding, and contextual knowledge to ensure the final output meets rigorous academic standards.

For developers of automated literature review technology, attention should be paid to the impact mechanisms of task context on system performance, fully considering the specific needs of different application scenarios during system design and optimization processes. Developers can provide intelligent parameter recommendation functions for users based on the performance boundary patterns revealed in this study, automatically matching the most suitable model and literature scale configurations according to user quality preferences. Additionally, developers should consider the trade-offs between different retrieval approaches: while self-built databases offer faster processing speeds, open domain retrieval systems provide higher literature relevance, though both approaches yield comparable content quality. Meanwhile, developers should also establish more refined evaluation systems, not only focusing on comparisons between systems and baseline methods, but also systematically testing system adaptability performance under different task conditions to provide users with more transparent and reliable technical solutions.

For technology standardization bodies, the strong performance of SocLitGen—surpassing graduate student writing across multiple quality dimensions—serves as an important cautionary signal. Such capability may incentivize over-reliance among graduate students, potentially undermining the cognitive development that manual literature review is designed to foster. Standardization bodies should therefore consider establishing usage guidelines that explicitly define appropriate boundaries for automated literature review tools in academic training contexts, distinguishing between assistive and substitutive use. Furthermore, when deploying such tools in real-world academic workflows, a human-machine collaborative framework should be mandated rather than fully autonomous operation: human judgment should be actively incorporated at critical decision nodes, including topic planning, literature screening, and synthesis evaluation, to ensure that automation serves as a cognitive scaffold rather than a replacement for scholarly reasoning.

8. Conclusion

Automated literature review generation technology has gradually become an important means for improving academic research efficiency. However, existing research is mainly concentrated in natural science and engineering fields such as computer science and chemistry, with relatively blank applications in social sciences. To address these issues, this study proposes an automated literature review generation framework specifically designed for social sciences—SocLitGen. The framework consists of seven modules: literature preprocessing, topic planning, literature collection, review generation, table generation, overview and commentary generation, and citation formatting, with adaptations in literature preprocessing and review generation specifically designed to meet social science requirements.

Based on a literature database of 1559,141 papers, experimental results demonstrate the SocLitGen framework's superior performance. Baseline comparisons show that in citation quality, SocLitGen outperforms Naive RAG by 39.9 % in recall and 70.1 % in precision, while achieving comparable performance to AutoSurvey. In content quality, SocLitGen demonstrates consistent advantages: SocLitGen surpasses AutoSurvey by 13.1 %, 18.5 %, and 18.7 % in academic professionalism, structural rationality, and language fluency, respectively; surpasses graduate student writing by 15.5 %, 9.5 %, and 25.4 % in the same dimensions. These results validate the effectiveness of domain-specific framework design for social science literature reviews. Cross-context evaluations further reveal differentiated performance patterns that provide practical deployment guidance: cross-disciplinary analysis confirms consistent performance across management and economics; cross-model experiments show Qwen2.5 performs optimally in academic and structural dimensions while GLM-4 excels in language fluency; cross-literature scale analysis demonstrates that large-scale sets enhance professionalism and structure but reduce fluency.

Although this study has achieved positive results, several limitations warrant attention in future research. First, experimental validation was primarily based on economics and management literature. Different social science disciplines exhibit distinct review conventions: psychological reviews emphasize measurement instruments and construct operationalizations, while sociological reviews

focus on competing theoretical explanations. This suggests that the framework's applicability in other sub-disciplines requires further verification, with discipline-specific adaptations needed to enhance universality. Second, the framework was evaluated exclusively using Chinese-language reviews, and LLM performance may vary across languages due to differences in training data, linguistic structures, and academic writing conventions; future research should assess cross-linguistic applicability. Third, general-purpose LLMs' limited domain-specific knowledge occasionally produces suboptimal outputs, such as overlapping sub-topics in the planning module and insufficient screening rigor in the collection module (see Appendix C); replacing these with domain-specific LLMs fine-tuned on social science literature represents a promising direction. Fourth, the framework targets reviews around predetermined topics and is less suited for exploratory scenarios where researchers scan broad domains to identify research gaps without predefined focal questions (Jiang, 2014), suggesting the need for complementary exploratory modules. Fifth, the current evaluation relies predominantly on quantitative rubric scores and citation counts, without qualitative assessment of review quality. Consequently, certain deficiencies beyond the defined metrics, such as critical depth and substantive informational density, may remain undetected. Future research should complement quantitative measures with qualitative assessment, including open-ended evaluator commentary to reveal multi-dimensional characteristics, direct comparisons of excerpts from generated and human-written reviews to provide concrete illustrations of their differences, and targeted interviews focused specifically on the shortcomings of generated reviews relative to human-written ones to identify concrete directions for improvement.¹⁰

Author statements

The authors declare no competing interest.

CRediT authorship contribution statement

Saiquan Hu: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Yushan Wei:** Writing – original draft, Software, Methodology. **Shuyang Sun:** Writing – original draft, Software, Formal analysis, Data curation.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ipm.2026.104885](https://doi.org/10.1016/j.ipm.2026.104885).

Data availability

Data will be made available on request.

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